

NOVEMBER 2026

Product Portfolio & Profitability Analysis

ANALYSIS ON BIKE SALES
IN A FEW COUNTRIES

11/11/2026



Introduction

SUMMARY

Graduated from President University majoring in International Relations with a broad understanding of global affairs, Indonesian politics, gender issues, and popular culture. A self-motivated self-starter that has improved his expertise and grown professionally. Highly interested in Data Analyst, with skills in data processing, building data pipelines, and big data analysis. Proficient in Python, SQL, and tools such as Tableau and PowerBI. Known for being detail-oriented, adaptable, and capable of working both independently and collaboratively. Ready to leverage prior work experience to support data management and optimization, contributing to data-driven decision-making.

WORK EXPERIENCES

Dinas Kesehatan Kota Bandung **July - November 2021**

Public Relations Staff

- Documenting and promoting an event created for the distribution of the Covid-19 vaccines
- Managed and organized participant data to ensure accuracy, consistency, and easy retrieval.
- Utilized Microsoft Excel or Google Sheets to organize, update, and manage data systematically.

EDUCATION

Dibimbing Bootcamp as Data Analyst **2026 - Present**

- Studied data storytelling, visualization, and performance measurement using industry tools.
- Technical Skills: Excel (pivot tables, data cleaning), SQL, Tableau, Power BI, Google Collab.
- Capstone Project: Designed an interactive dashboard to analyze company performance metrics and KPIs.

President University / International Relations / GPA (3.49 / 4.00) **2018 - 2022**

Bike Sales

Projects

Project Details



Objective

To compare sales performance across regions and identify the main contributors to sales and profit.

Purpose

My International Relations background taught me to understand regional differences and analyze issues from different perspectives. I applied this knowledge to compare sales across regions and understand how product categories and customer segments contribute to business performance.

Outcome

- The West region had the highest sales, followed by the East region.
- Technology was the top-selling product category, while the Consumer segment contributed the most sales.
- The analysis showed that higher sales do not always mean higher profit.
- This project helped me apply regional analysis skills to business data and develop data-driven recommendations.

Repository Superstore Analysis (Regional Sales Comparison)

Superstore Analysis using Python and Tableau

Project Details



Objective

To analyze marketing channel performance, customer behavior, and conversion rates to identify opportunities to improve marketing results.

Purpose

My International Relations studies helped me develop communication and analytical skills. My previous experience in Public Relations also taught me the importance of understanding audiences. These experiences motivated me to analyze customer behavior and understand how marketing channels influence purchasing decisions.

Outcome

- Paid Ads generated the highest number of purchases, followed by Social Media and Organic Search.
- Instagram showed the largest drop-off between impressions and clicks.
- The analysis identified opportunities to improve advertising, customer conversion, and repeat purchases.
- This project helped me combine my communication background with data analysis to support marketing decisions.

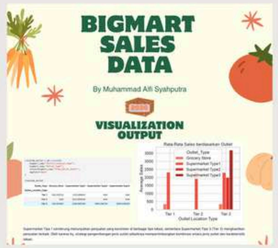
Repository Marketing Growth (Customer Journey)

Marketing Growth Analysis using Python and Tableau



Olist E-commerce Analysis using SQL

Project Details



Objective

To analyze BigMart sales data and identify factors related to product sales, pricing, and outlet performance using Python.

Purpose

As an International Relations graduate, I learned to think critically, analyze problems, and understand relationships between different factors. I applied these skills to sales data to understand business performance and identify opportunities for improvement.

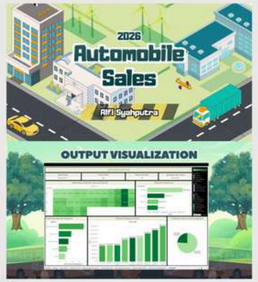
Outcome

- Tier 2 outlets showed strong average sales, while Tier 3 outlets contributed significantly to overall sales.
- The analysis identified opportunities to improve product promotions and inventory management.
- This project helped me apply my analytical thinking to business problems and develop my skills as a Data Analyst.

Repository Bigmart Sales Data

Bigmart Sales Data Analysis using Python and Tableau

Project Details



Objective

To analyze automobile sales performance, identify the best-selling products and purchasing patterns, and evaluate customer behavior to understand sales trends and potential customer churn risks.

Purpose

My International Relations degree trained me to understand social dynamics, behavioral patterns, and relationships between different groups. This background encouraged me to explore customer behavior through data analysis. In this project, I used Python and Tableau to clean, process, analyze, and visualize automobile sales data. I applied sales trend analysis, product performance analysis, and RFM customer segmentation to transform raw transaction data into actionable business insights.

Outcome

- Sales increased by 39%, from approximately $3.35 million in 2018 to $4.67 million in 2019. Sales from January to May 2020 reached approximately $1.74 million.
- Classic Cars was the best-selling product line, generating approximately $822,000 in sales, nearly twice the revenue of Vintage Cars ($398,000).
- Customer segmentation identified 9 At Risk customers, representing approximately 10% of customers but contributing around $1.08 million (11% of total sales). This highlights the potential revenue impact of customer churn.
- Recommendations: prepare sales campaigns ahead of the November sales peak, maintain Classic Cars as a key product while promoting other product lines, and implement targeted retention strategies for Champions, Loyal, Potential Loyalists, and At Risk customers.

Repository Automobile-Analysis

Automobile Sales Analysis using Python and Tableau

Journal

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ISSN 2723-6250 (online)
DOI: https://doi.org/10.33122/ejeset.v6i2.698

SAINTIS publishing

Research Article

Bike Sales Analysis for Understanding Market Trends in Europe via Power BI Dashboard

Fitri Nur Waliyaden[1], Gloria Theodora Wahi Leo[1], Indriana Kurnia Cahyati[1], Mery Adelia[1], Akhmad Bakhrun[2]*

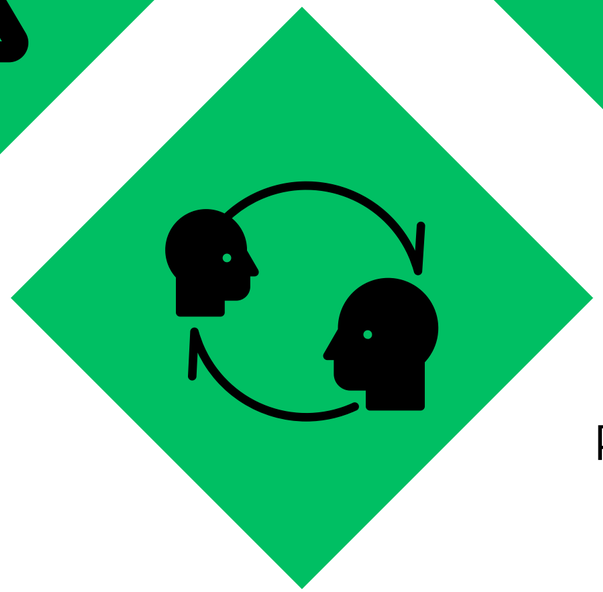
The rapid development of information technology in the industrial era 4.0 has brought fundamental changes in business data management. Sales data, which was once merely a record of transactions, is now positioned as a strategic asset with the potential to provide deep insights into consumer behavior, market segmentation, and the effectiveness of sales strategies. This transformation compels companies to develop advanced data analysis capabilities in order to make adaptive, accurate, and evidence-based decisions. In this landscape, Business Intelligence (BI) systems have emerged as transformative tools, enabling organizations to process vast amounts of data and convert it into actionable insights that improve decision-making and operational performance through real-time monitoring, predictive analytics, and automated alerts (Putra et al., 2025).

https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.ejeset.saintispub.com/ejeset/article/download/698/506&ved=2ahUKEwiDkYLijP2XAxUwRCoJHU7xHoUQFnoECBwQAQ&usg=AOvVaw0YHRUi8CPIgztB4pdtDT_h

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Problem Statement



1. Kategori produk manakah yang menghasilkan margin keuntungan tertinggi?



2. Bagaimana profitabilitas produk bervariasi di berbagai negara?



3. Kelompok usia pelanggan manakah yang menghasilkan margin keuntungan tertinggi, dan bagaimana perusahaan dapat mengoptimalkan strategi produknya untuk setiap segmen tersebut?



	Date	Day	Month	Year	Customer_Age	Age_Group	Customer_Gender	Country	State	Product_Category	Sub_Category	Product	Order_Quantity	Unit_Cost	Unit_Price	Profit	Cost	Revenue
0	2013-11-26	26	November	2013	19	Youth (<25)	M	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45	120	590	360	950
1	2015-11-26	26	November	2015	19	Youth (<25)	M	Canada	British Columbia	Accessories	Bike Racks	Hitch Rack - 4-Bike	8	45	120	590	360	950
2	2014-03-23	23	March	2014	49	Adults (35-64)	M	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	23	45	120	1366	1035	2401
3	2016-03-23	23	March	2016	49	Adults (35-64)	M	Australia	New South Wales	Accessories	Bike Racks	Hitch Rack - 4-Bike	20	45	120	1188	900	2088

```
Date                str
Day               int64
Month               str
Year              int64
Customer_Age      int64
Age_Group           str
Customer_Gender     str
Country             str
State               str
Product_Category    str
Sub_Category        str
Product             str
Order_Quantity    int64
Unit_Cost         int64
Unit_Price        int64
Profit            int64
Cost              int64
Revenue           int64
dtype: object
```

```
data = pd.read_csv("Bike Sales in Europe.csv")
```

Dataset berisi **113.036 baris** transaksi dan **18 kolom**. **Sembilan kolom berupa teks (Date, Month, Age_Group, Customer_Gender, Country, State, Product_Category, Sub_Category, Product)** dan **sembilan kolom berupa angka**. Kolom **Date** masih **berupa teks**, jadi perlu **diubah** ke **format tanggal** sebelum dianalisis berdasarkan waktu.

Sumber data Kaggle: Bike Sales

DATA UNDERSTANDING

No	Kolom	Tipe	Deskripsi	Isi / rentang
1	`Date`	Teks (tanggal)	Tanggal transaksi, format `YYYY-MM-DD`.	2011-01-01 sampai 2016-07-31
2	`Day`	Angka	Hari (tanggal) dalam bulan, turunan dari `Date`.	1 sampai 31
3	`Month`	Teks	Nama bulan, turunan dari `Date`.	January sampai December
4	`Year`	Angka	Tahun transaksi, turunan dari `Date`.	2011 sampai 2016
5	`Customer_Age`	Angka	Usia pelanggan saat transaksi, dalam tahun.	17 sampai 87 (median 35)
6	`Age_Group`	Teks	Kelompok usia pelanggan, turunan dari `Customer_Age`.	Youth (<25): 17-24, Young Adults (25-34), Adults (35-64), Seniors (64+): 65-87
7	`Customer_Gender`	Teks	Jenis kelamin pelanggan.	M (58.312), F (54.724)
8	`Country`	Teks	Negara tempat transaksi terjadi.	6 negara: United States, Australia, Canada, United Kingdom, Germany, France
9	`State`	Teks	Negara bagian atau provinsi di dalam `Country`	53 wilayah (US 22, France 16, Germany 6, Australia 5, Canada 3, UK 1)
10	`Product_Category`	Teks	Kategori produk tingkat teratas.	Accessories (70.120), Bikes (25.982), Clothing (16.934)
11	`Sub_Category`	Teks	Subkategori di dalam kategori produk.	17 subkategori (Accessories 8, Clothing 6, Bikes 3)
12	`Product`	Teks	Nama produk spesifik, termasuk varian warna atau ukuran.	130 produk, misalnya “Hitch Rack - 4-Bike”
13	`Order_Quantity`	Angka	Jumlah unit yang dibeli dalam satu transaksi.	1 sampai 32 (median 10)
14	`Unit_Cost`	Angka	Biaya pokok per satu unit.	1 sampai 2.171
15	`Unit_Price`	Angka	Harga jual per satu unit.	2 sampai 3.578
16	`Cost`	Angka	Total biaya transaksi, yaitu `Order_Quantity × Unit_Cost`.	1 sampai 42.978
17	`Revenue`	Angka	Total pendapatan transaksi.	2 sampai 58.074 (median 223)
18	`Profit`	Angka	Keuntungan transaksi, yaitu `Revenue − Cost`.	-30 sampai 15.096






COLUMN DESCRIPTION

```python
inkonsistensi = df_copy.select_dtypes(include=["str"]).columns # Mengambil nama kolom dengan tipe data string

for i in inkonsistensi : # Memeriksa inkonsistensi data pada kolom dengan tipe data string
    print(f"Kolom {i} :")
    print(df_copy[i].unique())
    print("\n")
```

```
Kolom Sub_Category :
<ArrowStringArray>
[       'Bike Racks',       'Bike Stands', 'Bottles and Cages',
             'Caps',          'Cleaners',           'Fenders',
          'Gloves',           'Helmets',   'Hydration Packs',
         'Jerseys',    'Mountain Bikes',        'Road Bikes',
          'Shorts',             'Socks',   'Tires and Tubes',
     'Touring Bikes',             'Vests']
Length: 17, dtype: str


Kolom Product :
<ArrowStringArray>
[    'Hitch Rack - 4-Bike',  'All-Purpose Bike Stand',
    'Mountain Bottle Cage',   'Water Bottle - 30 oz.',
        'Road Bottle Cage',            'AWC Logo Cap',
   'Bike Wash - Dissolver',   'Fender Set - Mountain',
   'Half-Finger Gloves, L',   'Half-Finger Gloves, M',
 ...
 'Touring-1000 Yellow, 54',   'Touring-2000 Blue, 54',
   'Touring-3000 Blue, 50', 'Touring-3000 Yellow, 58',
   'Touring-2000 Blue, 46',   'Touring-2000 Blue, 50',
   'Touring-2000 Blue, 60',         'Classic Vest, L',
         'Classic Vest, M',         'Classic Vest, S']
Length: 130, dtype: str


Kolom Customer_Gender :
<ArrowStringArray>
['M', 'F']
Length: 2, dtype: str


Kolom Country :
<ArrowStringArray>
['Canada', 'Australia', 'United States', 'Germany', 'France',
 'United Kingdom']
Length: 6, dtype: str
```

```python
for i in inkonsistensi:
    print(i, ":", df_copy[i].str.strip().ne(df_copy[i]).sum())
```

```
Date : 0
Month : 0
Age_Group : 0
Customer_Gender : 0
Country : 0
State : 0
Product_Category : 0
Sub_Category : 0
Product : 0
```

```python
for i in inkonsistensi:
    df_copy[i] = df_copy[i].str.strip()
```

Penulisan data sudah rapi dan tidak ada spasi berlebih

DATA INCONSISTENCY

```python
df_copy_miss = df_copy.isnull().sum() # Mencari jumlah missing value
df_copy_miss_percentage = df_copy_miss / len(df_copy) * 100 # Mencari persentase missing value

print(f"Jumlah missing value : \n{df_copy_miss}")
print(f"Persentase missing value : \n{df_copy_miss_percentage}")
```

```
Jumlah missing value :
Date                0
Day                 0
Month               0
Year                0
Customer_Age        0
Age_Group           0
Customer_Gender     0
Country             0
State               0
Product_Category    0
Sub_Category        0
Product             0
Order_Quantity      0
Unit_Cost           0
Unit_Price          0
Profit              0
Cost                0
Revenue             0
dtype: int64
```

```
Persentase missing value
Date                0.0
Day                 0.0
Month               0.0
Year                0.0
Customer_Age        0.0
Age_Group           0.0
Customer_Gender     0.0
Country             0.0
State               0.0
Product_Category    0.0
Sub_Category        0.0
Product             0.0
Order_Quantity      0.0
Unit_Cost           0.0
Unit_Price          0.0
Profit              0.0
Cost                0.0
Revenue             0.0
dtype: float64
```

Tidak ada missing value di seluruh 18 kolom (0 baris atau 0%). Data sudah lengkap, jadi tidak perlu mengisi nilai kosong ataupun ditangani.

MISSING VALUE

Duplicate Value (Jumlah dan Persentase)

```
duplicate_all = df_copy.duplicated().sum() # Memeriksa jumlah duplicate value berdasarkan tiap baris
persent_duplicate_all = duplicate_all /len(df_copy)*100 # Menghitung persentase duplicate value
print(f"Jumlah Baris yang duplikat = {duplicate_all}")
print(f"Persentase Baris yang duplikat = {persent_duplicate_all:.2f}%")
```

```
Jumlah Baris yang duplikat = 1000
Persentase Baris yang duplikat = 0.88%
```

Menampilkan baris yang duplicate

```
dup_all = df_copy[df_copy.duplicated(keep=False)]
dup_all.sort_values(by=list(df_copy.columns)).head(10)
```

	Date	Day	Month	Year	Customer_Age	Age_Group	Customer_Gender	Country	State	Product_Category	Sub_Category	Product
69018	2011-02-20	20	February	2011	25	Young Adults (25-34)	F	Canada	British Columbia	Bikes	Road Bikes	Road-250 Red, 52
69019	2011-02-20	20	February	2011	25	Young Adults (25-34)	F	Canada	British Columbia	Bikes	Road Bikes	Road-250 Red, 52
58238	2011-03-03	3	March	2011	43	Adults (35-64)	F	United States	California	Bikes	Road Bikes	Road-150 Red, 44
58239	2011-03-03	3	March	2011	43	Adults (35-64)	F	United States	California	Bikes	Road Bikes	Road-150 Red, 44

DUPLICATE

Ada **1.000 baris duplikat (0,88%)** dari 113.036 baris. Artinya seluruh 18 kolom pada baris tersebut sama persis dengan baris lain. Jumlahnya kecil, tetapi tetap perlu ditangani agar Revenue dan Profit tidak terhitung dua kali.

Pada 4 pasang baris yang ditampilkan, kedua baris kembarnya **berurutan** (misalnya baris 69018 dan 69019, atau 58238 dan 58239) dengan tanggal, usia, negara, produk, jumlah, harga, dan revenue yang sama persis. Baris kembar yang saling bersebelahan seperti ini lebih mirip **input ganda** daripada dua pembelian terpisah, karena dua pembelian terpisah jarang tercatat tepat berurutan dengan data identik. Itulah alasan baris tersebut dihapus dan hanya satu salinan yang disimpan.

OUTLIER

```
cek_outlier = df_copy.select_dtypes(include=["int64"]).columns
cek_outlier
```

```
Index(['Day', 'Year', 'Customer_Age', 'Order_Quantity', 'Unit_Cost',
       'Unit_Price', 'Profit', 'Cost', 'Revenue', 'Date_code', 'Month_code',
       'Age_Group_code', 'Customer_Gender_code', 'Country_code', 'State_code',
       'Product_Category_code', 'Sub_Category_code', 'Product_code'],
      dtype='str')
```

	Kolom	Jumlah Outlier	Persentase (%)
4	Unit_Cost	25984	22.987367
5	Unit_Price	25984	22.987367
7	Cost	18179	16.082487
8	Revenue	14765	13.062210
6	Profit	10021	8.865317
9	Date_code	1294	1.144768
2	Customer_Age	534	0.472416

```
summary_data = []
kolom_dengan_outlier = []

for i in cek_outlier:
    Q1 = df_copy[i].quantile(0.25)
    Q3 = df_copy[i].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outlier = (df_copy[i] < lower) | (df_copy[i] > upper)
    jumlah_outlier = outlier.sum()
    persentase_outlier = (jumlah_outlier / len(outlier)) * 100

    # Simpan data ke dalam list untuk dibuat DataFrame nanti
    summary_data.append({
        'Kolom': i,
        'Jumlah Outlier': jumlah_outlier,
        'Persentase (%)': persentase_outlier
    })

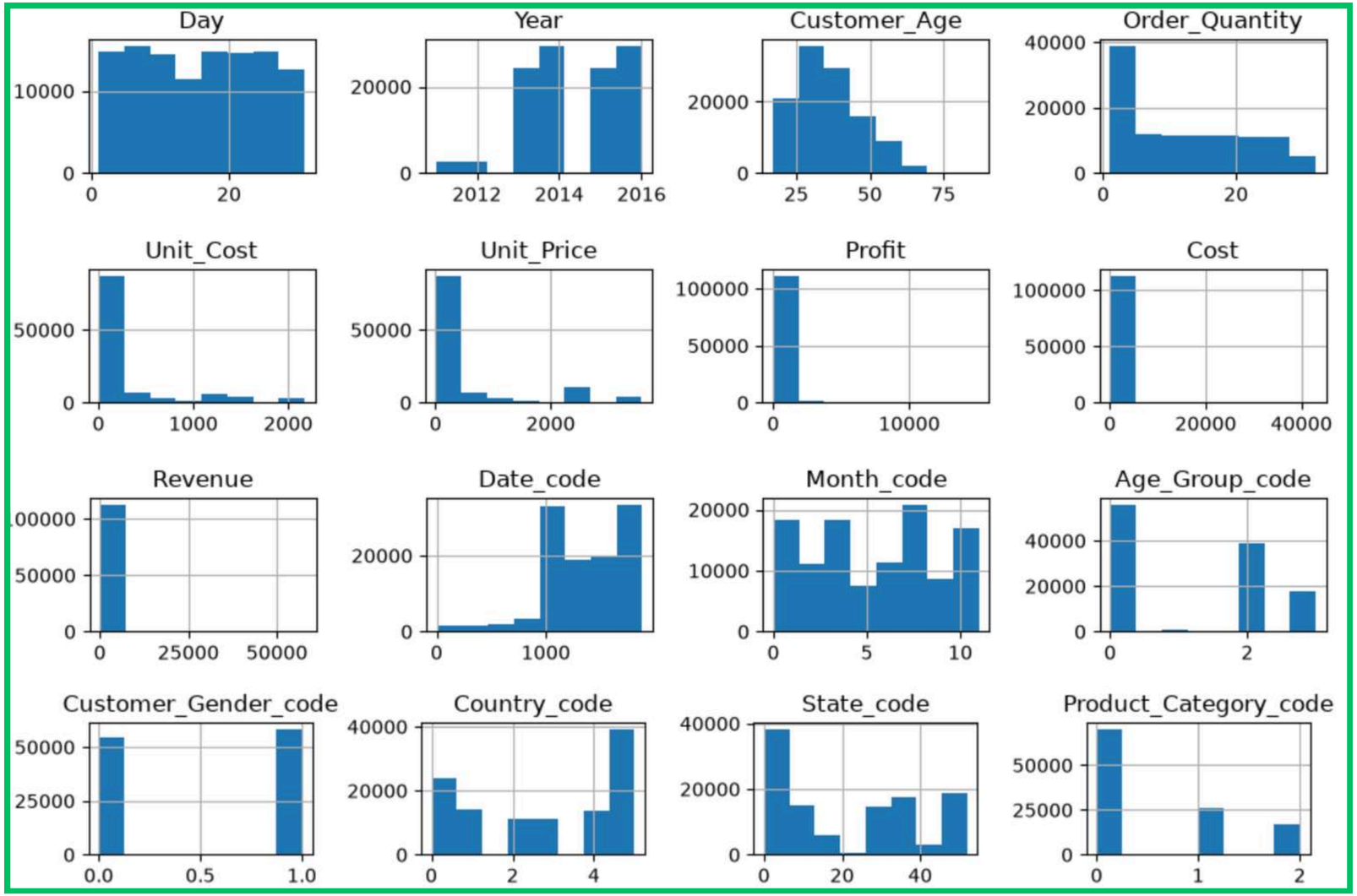
    # Cek apakah kolom ini memiliki outlier (> 0)
    if jumlah_outlier > 0:
        kolom_dengan_outlier.append(df_copy[i])

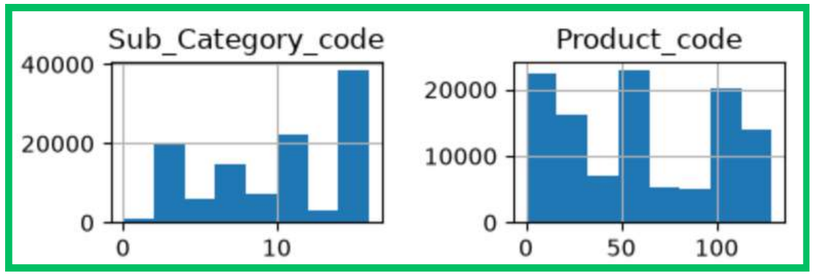
df_outlier_summary = pd.DataFrame(summary_data).sort_values(by=("Jumlah Outlier"), ascending=False)

display(df_outlier_summary)
```

Terdapat Outlier terhadap variabel numerik, seperti harga barang (Unit price), biaya produksi (Unit cost), dan umur (age), secara alami dapat memiliki rentang nilai yang cukup luas. Perbedaan karakteristik produk, skala produksi, segmentasi pasar, serta kondisi individu dapat menyebabkan variasi nilai yang signifikan.

DATA DISTRIBUTION





Berdasarkan data distribusi tersebut, aktivitas transaksi selama 2013–2016 relatif stabil dan tersebar merata sepanjang bulan. Mayoritas pelanggan berada pada rentang usia produktif, khususnya Adults (35–64 tahun), dengan pelanggan laki-laki sedikit lebih dominan (52%). Dari sisi penjualan, sebagian besar pelanggan melakukan pembelian dalam jumlah kecil (1–4 unit), dengan produk berharga rendah mendominasi transaksi (77%). Accessories menjadi kategori produk paling diminati, disusul Bikes dan Clothing. Sementara itu, Profit, Cost, dan Revenue menunjukkan distribusi right-skewed, yang berarti sebagian besar transaksi bernilai kecil, tetapi terdapat sejumlah kecil transaksi dengan nilai tinggi. United States dan Australia menjadi pasar dengan jumlah transaksi terbesar. Untuk mengoptimalkan penjualan, perusahaan disarankan mempertahankan pelanggan utama melalui program loyalitas, meningkatkan jumlah pembelian melalui strategi bundling dan cross-selling, menjaga ketersediaan stok produk terlaris, serta mengembangkan pemasaran berdasarkan karakteristik usia dan negara.

DATA PREPARATION

Feature Engineering

Penanganan menghitung ulang

```
df_copy = df_copy.drop(['Profit', 'Cost', 'Revenue'], axis =1)
```

```
df_copy['Total_Cost'] = df_copy['Order_Quantity'] * df_copy['Unit_Cost']
df_copy['Revenue'] = df_copy['Order_Quantity'] * df_copy['Unit_Price']
df_copy['Profit'] = df_copy['Revenue'] - df_copy['Total_Cost']
```

```
df_copy[["Profit", "Total_Cost", "Revenue"]].head(5)
```

	Profit	Total_Cost	Revenue
0	600	360	960
1	600	360	960
2	1725	1035	2760
3	1500	900	2400
4	300	180	480

Penanganan duplicate

```
df_copy = df_copy.drop_duplicates()
df_copy.duplicated().sum()
```

```
np.int64(0)
```

```
print(f"Jumlah duplicate : {df_copy.duplicated().sum()}")
print(f"Persentase duplicate : {df_copy.duplicated().mean() * 100:.2f}%")
```

```
Jumlah duplicate : 0
Persentase duplicate : 0.00%
```

Penanganan Datetime

```
df_copy["Date"] = pd.to_datetime(df_copy["Date"], errors="coerce")
```

```
df_copy.dtypes
```

```
Date                          datetime64[us]
```

Kondisi Data Setelah Data Preparation & Cleaning:

- Data siap dianalisis: 112.036 baris, 0 missing value, 0 duplikat, dan semua tanggal valid.
- Label teks konsisten (tidak ada spasi berlebih atau label ganda).
- Revenue dan Profit dihitung ulang (Total_Cost = Order_Quantity x Unit_Cost, Revenue = Order_Quantity x Unit_Price, Profit = Revenue - Total_Cost).
- Outlier pada kolom uang dipertahankan karena itu produk dan pesanan bernilai besar yang memang ada.

1.000 baris kembar sudah dihapus dan sisanya **0 duplikat**. Data tinggal **112.036 baris** (113.036 dikurangi 1.000). Karena yang dihapus hanya 0,88% dan sifatnya salinan identik, bentuk data secara keseluruhan tidak berubah.

Kruskal–Wallis: Hubungan Profit dengan Product Category

- **H0:** Tidak terdapat perbedaan distribusi Profit yang signifikan antara kategori Accessories, Bikes, dan Clothing.
- **H1:** Terdapat setidaknya satu kategori produk yang memiliki distribusi Profit berbeda secara signifikan.

EDA

```python
# Menentukan tingkat signifikansi
alpha = 0.05

# Mengelompokkan Profit berdasarkan Product_Category
data_profit = [
    grup["Profit"].dropna().values
    for nama, grup in df_copy.groupby("Product_Category")
]

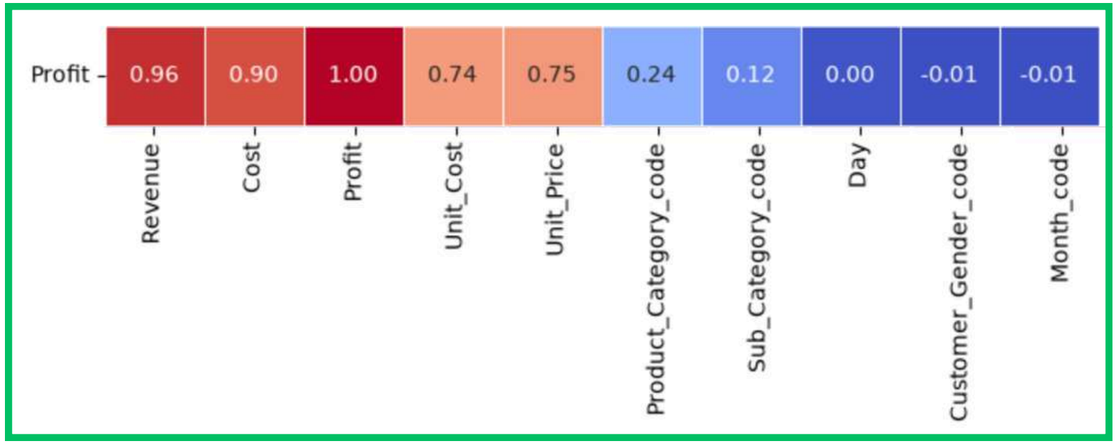
# Mengambil nama kategori produk
nama_kategori = df_copy["Product_Category"].dropna().unique().tolist()

# Menjalankan uji Kruskal-Wallis
h_stat, p_value = kruskal(*data_profit)

print("Product Category:", nama_kategori)
print("Kruskal-Wallis H:", h_stat)
print("P-value:", p_value)

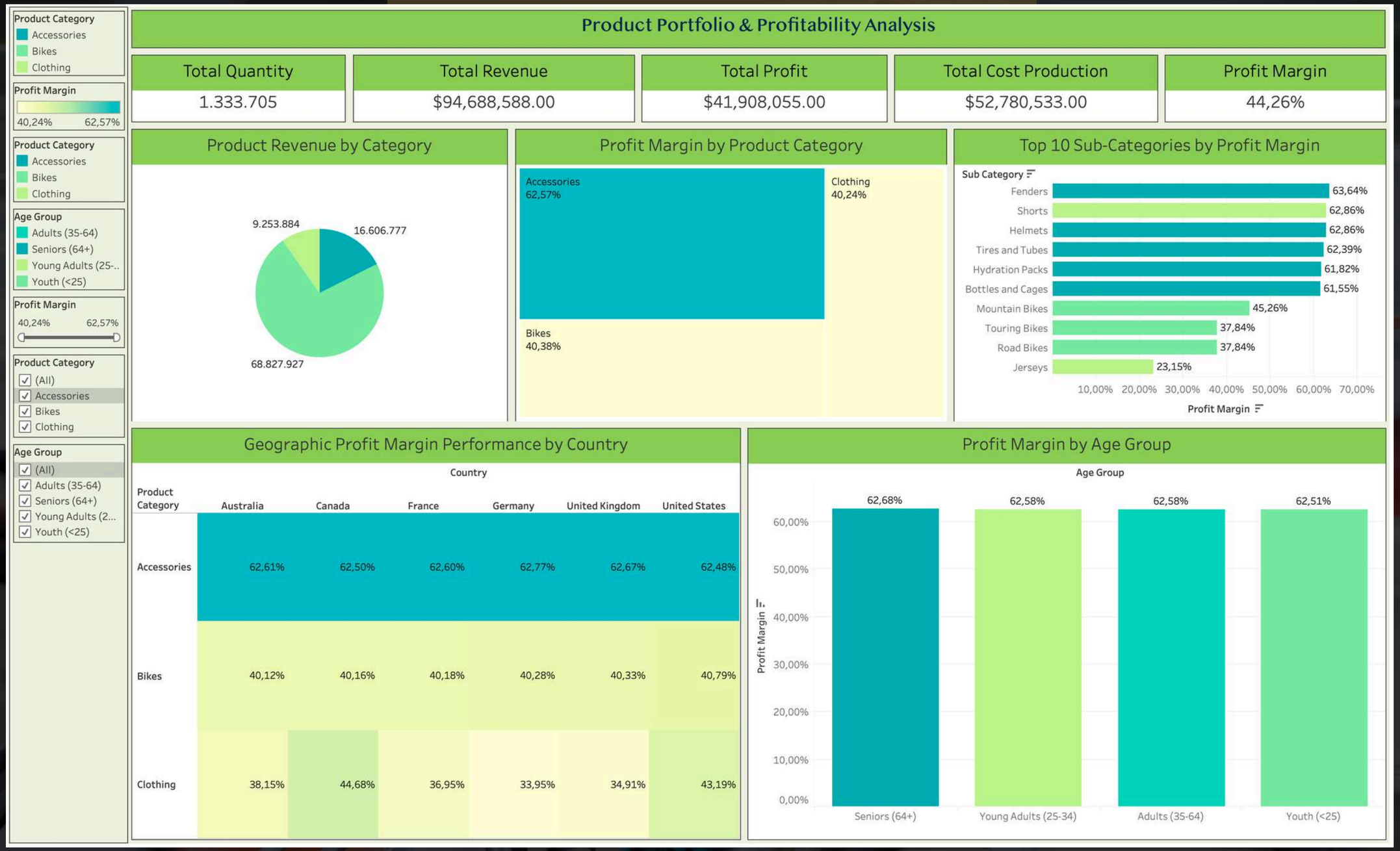
# Menentukan hasil pengujian
if p_value < alpha:
    print("Keputusan: Tolak H0.")
    print("Terdapat perbedaan distribusi Profit yang signifikan pada setidaknya dua kategori produk.")
else:
    print("Keputusan: Gagal menolak H0.")
    print("Belum terdapat bukti perbedaan distribusi Profit yang signifikan antar kategori produk.")
```

```
Product Category: ['Accessories', 'Clothing', 'Bikes']
Kruskal-Wallis H: 46566.325698130124
P-value: 0.0
Keputusan: Tolak H0.
Terdapat perbedaan distribusi Profit yang signifikan pada setidaknya dua kategori produk.
```

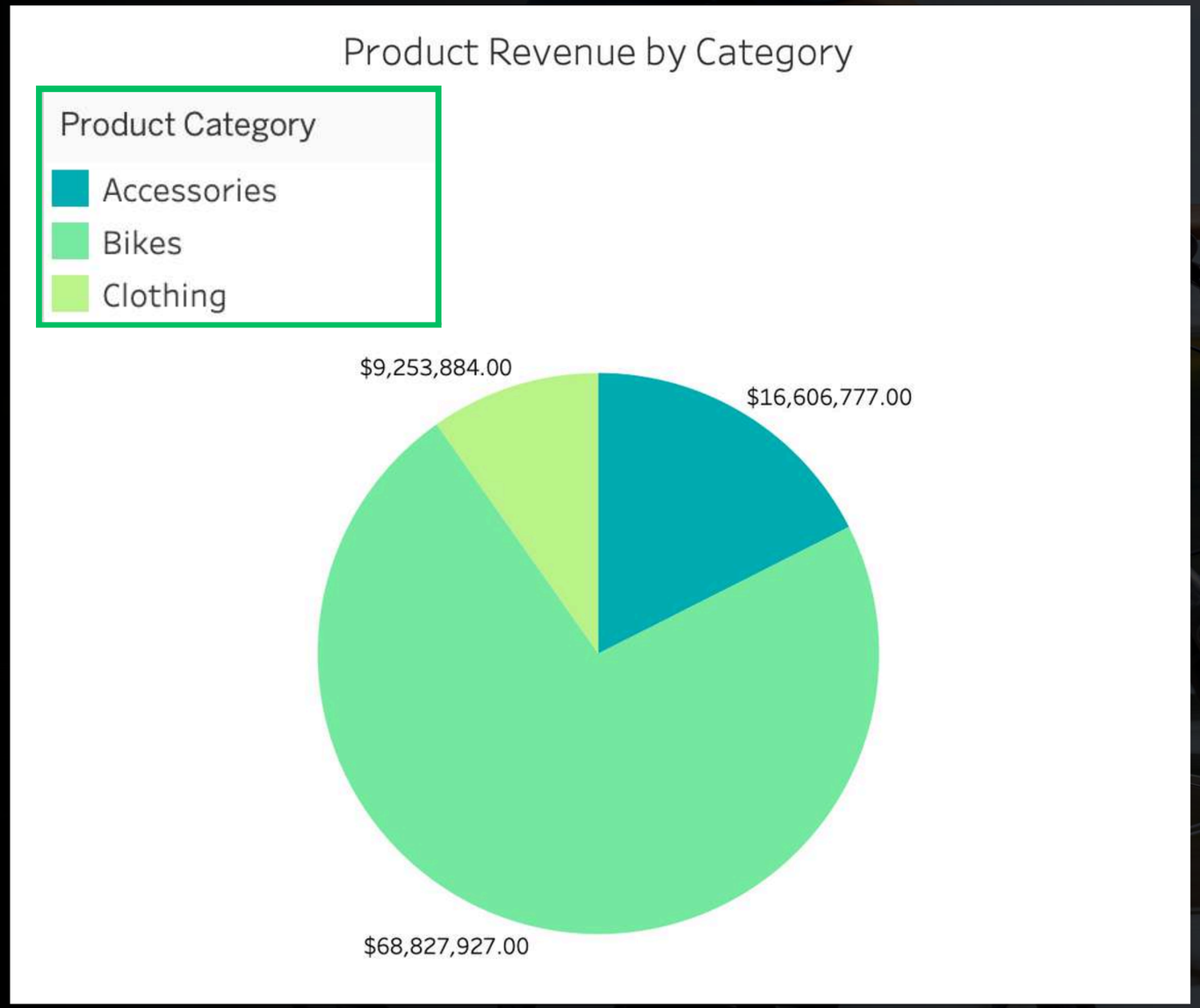


Nilai uji statistik Kruskal-Wallis sangat besar, yaitu 46.566,33. Hal ini menghasilkan p-value yang sangat kecil sehingga Python membulatkannya menjadi 0.0 ketika ditampilkan dalam format desimal biasa.

OUTPUT VISUALIZATION



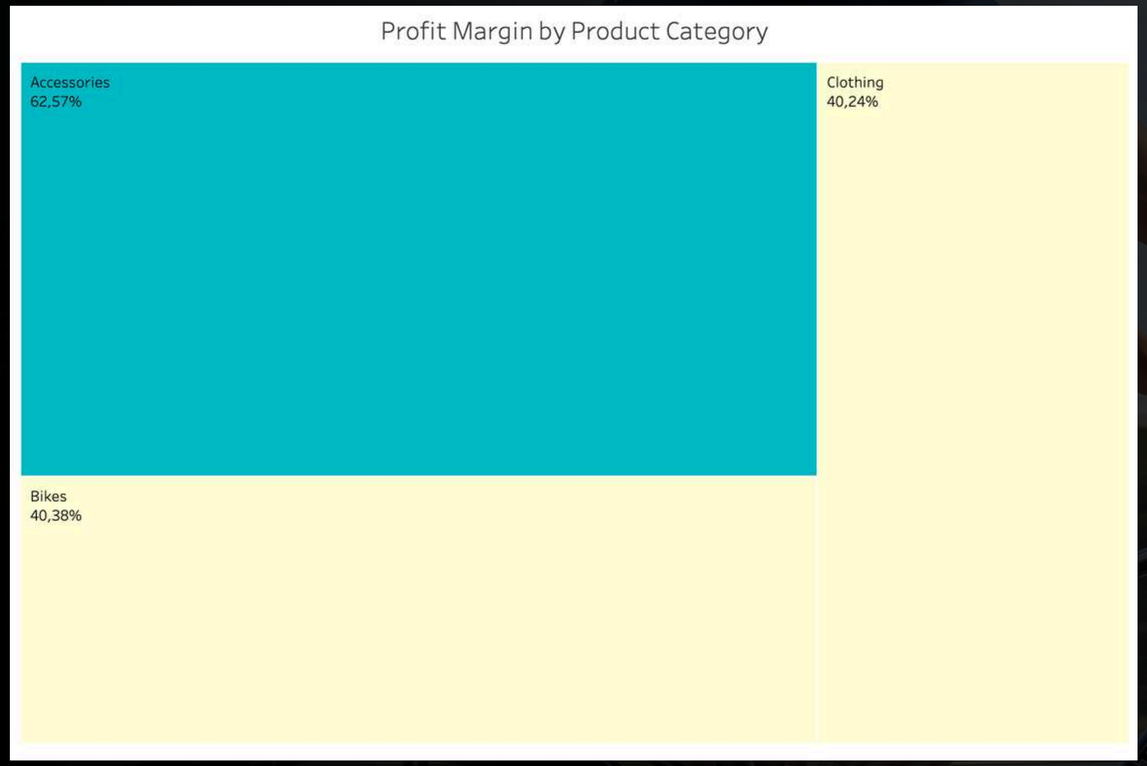
OUTPUT VISUALIZATION

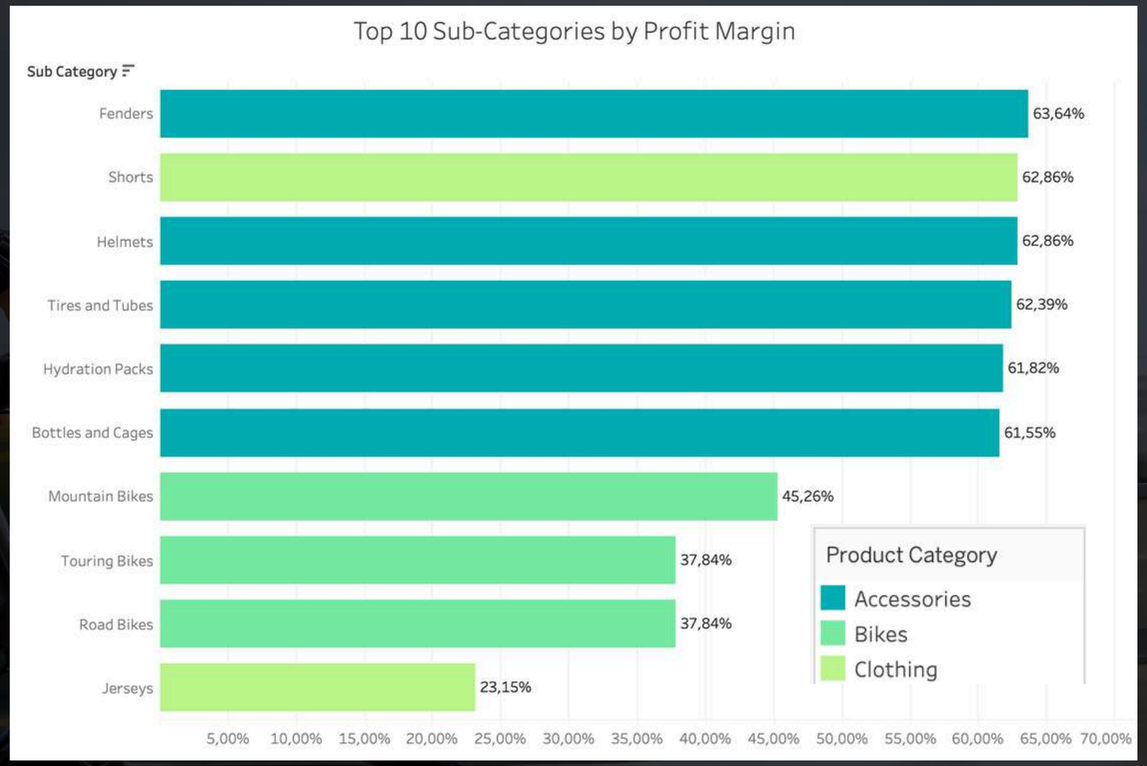


Kategori Bikes memiliki kontribusi Revenue paling besar, sekitar 72,69%,yang mengartikan adanya indikasi dominasi produk terlaris. Apabila hal ini dibiarkan maka perusahaan memiliki ketergantungan pemasukan dan kestabilan revenue yang diperoleh. Hal ini menjadi peluang untuk mempertahankan performa Bikes, tetapi juga menjadi pengingat bahwa perusahaan perlu mengembangkan kategori lain agar sumber pendapatannya lebih seimbang.

Namun hal ini perlu dilakukan pengecekan apakah penjualan terbesar menghasilkan profit yang besar?

OUTPUT VISUALIZATION





Namun, ketika kita melihat Profit Margin, kita menemukan perspektif yang berbeda. Accessories justru memiliki tingkat Profit Margin yang lebih tinggi. Hal ini menunjukkan bahwa strategi perusahaan tidak cukup hanya berfokus pada kategori barang yang paling banyak terjual. Perusahaan juga perlu memperhatikan kategori yang mampu menghasilkan keuntungan yang paling tinggi karena biaya tersebut menentukan keuntungan Perusahaan yang diperoleh

Dibuktikan dengan hasil Analisa Sub category dengan profit margin tertinggi masih dipegang category Accessories.

OUTPUT VISUALIZATION

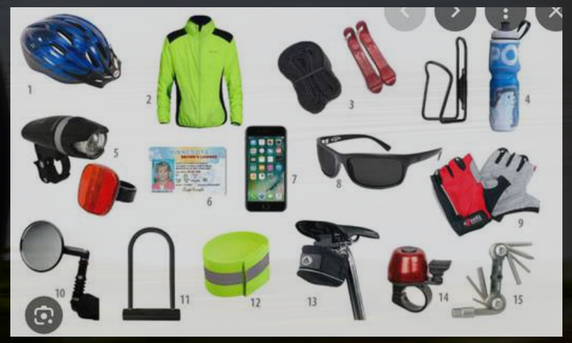
Geographic Profit Margin Performance by Country

Product Category	Australia	Canada	France	Germany	United Kingdom	United States
	Country					
Accessories	62,61%	62,50%	62,60%	62,77%	62,67%	62,48%
Bikes	40,12%	40,16%	40,18%	40,28%	40,33%	40,79%
Clothing	38,15%	44,68%	36,95%	33,95%	34,91%	43,19%

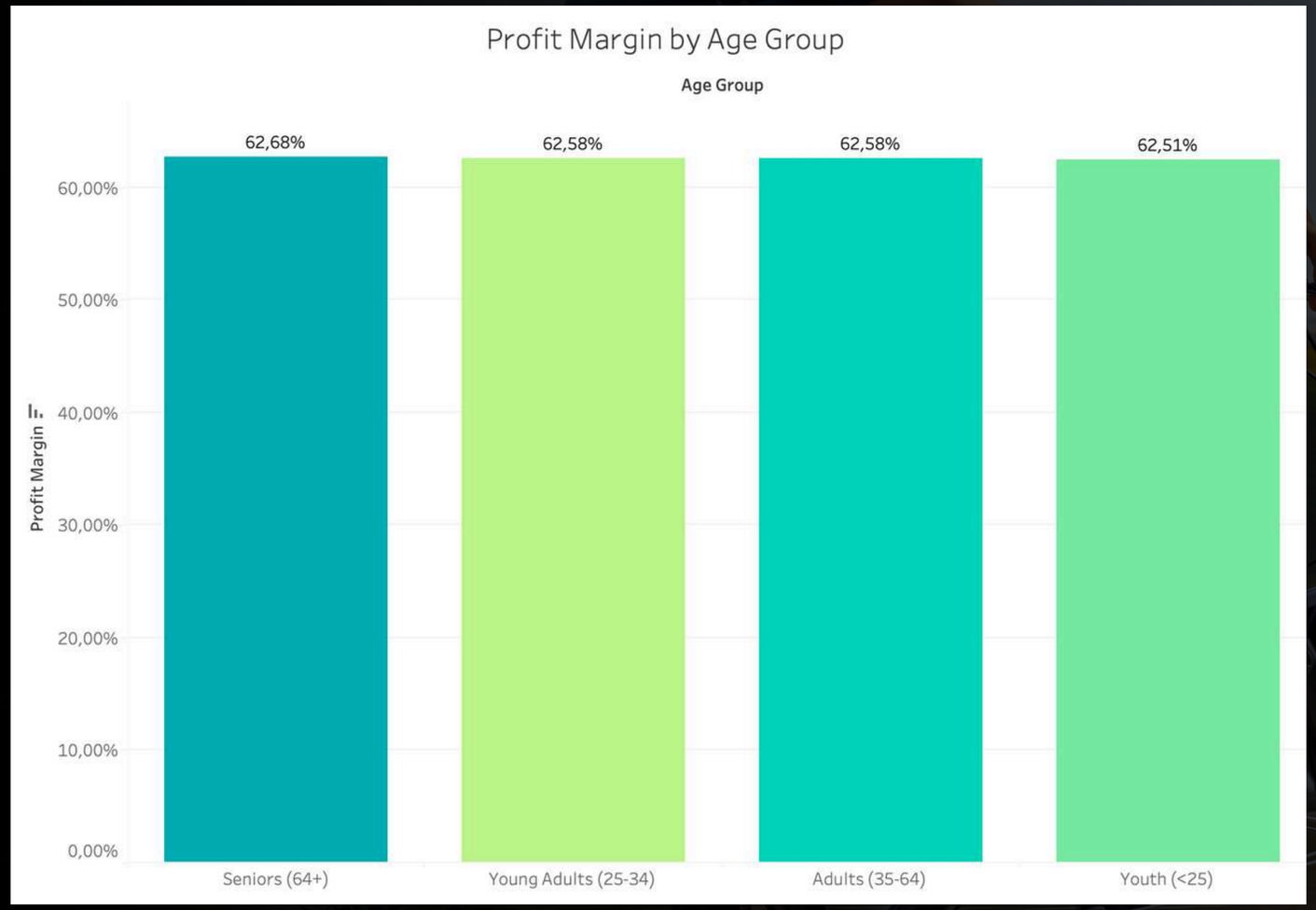
Accessories menunjukkan performa yang cukup stabil di seluruh negara, dengan Profit Margin berkisar antara 62,48% hingga 62,77%. Hal ini menunjukkan bahwa Accessories tidak hanya memiliki tingkat profitabilitas yang tinggi secara keseluruhan, tetapi juga mampu mempertahankan margin yang relatif konsisten di berbagai pasar.

Setelah melihat perbedaan profitabilitas berdasarkan negara, kita juga perlu memahami apakah karakteristik pelanggan memiliki hubungan dengan tingkat keuntungan yang dihasilkan





OUTPUT VISUALIZATION



Margin Accessories bersifat stabil dan tinggi di semua kelompok usia. Profitabilitasnya ditentukan oleh kategori produknya. Hal ini sejalan dengan temuan sebelumnya bahwa Accessories bermargin 62,57%, jauh di atas Bikes (40,38%) dan Clothing (40,24%).

Pertahankan satu strategi harga dan margin Accessories untuk semua usia. Margin yang seragam menunjukkan struktur harga dan biaya sudah sehat. Perusahaan bisa memanfaatkan Accessories sebagai penopang margin lewat bundling dengan sepeda

CONCLUSION AND RECOMMENDATIONS

- **Strategi Cross-Selling dan Bundle untuk Produk Sepeda**

Mengingat Bikes menyumbang volume penjualan terbesar namun memiliki margin lebih rendah dibandingkan aksesoris, perusahaan disarankan menerapkan strategi bundling atau cross-selling saat penjualan unit sepeda. Setiap pembelian unit sepeda (Bikes) dapat dipaketkan dengan Accessories bermargin tinggi (seperti Helmets, Fenders, atau Bottles and Cages)untuk mendongkrak total nilai transaksi sekaligus menaikkan rata-rata overall profit margin dari lini penjualan sepeda.

- **Ekspansi dan Maksimalkan Penjualan pada Demografis Berdaya Beli Tinggi**

Dengan tingkat profitabilitas yang konsisten stabil di angka 62.5% di seluruh kelompok umur, bisnis dapat meningkatkan anggaran pemasaran untuk memperluas jangkauan pasar ke kelompok Seniors dan Young Adults yang mencatatkan margin tertinggi. Strategi kemitraan, program loyalitas, serta kampanye iklan terstruktur dapat difokuskan pada segmen-segmen ini untuk memaksimalkan akumulasi keuntungan bersih secara berkelanjutan.

- **Skalasi Penjualan Sub-Kategori Unggulan dan Pasar Amerika Utara**

Perusahaan bisa memperbesar volume alokasi stok dan fokus promosi pada sub-kategori bermargin tertinggi seperti Fenders, Shorts, dan Helmets (margin >62%).



BikeSales

Thank You

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https://github.com/Alfi-Syahputra/Bike-Sales-Analysis